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International factors and inflation risks

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ABSTRACT

Assessing the risks of having either very low or very high inflation is crucial for policymakers, businesses, and householders. In a globalised world, these risks are increasingly determined by international conditions. In this paper, we empirically analyse the impact of international inflation factors on forecasting monthly domestic inflation risks in a large number of economies observed worldwide from 1999 to 2022. Risk forecasts are obtained using factor-augmented quantile regressions estimated with international factors extracted from a multi-level dynamic factor model with overlapping blocks of inflations corresponding to economies grouped either in a given geographical region or according to their development level. We conclude that in a large number of countries, international factors are relevant to explain the right tail of the distribution of inflation, and consequently they are more relevant for the risk related to high inflation than for low inflation. The role of international factors is stronger in developed European countries, while the inflation risks of low-income developing countries are hardly affected by international conditions, and the results for middle-income countries are mixed. We also show that the predictive power of international factors has increased in the most recent years of high inflation.

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1. Introduction

In this paper, we investigate whether international factors help forecast the upside and downside risks of domestic inflation, a question that remains unanswered despite its clear policy relevance. Much of the vast literature on modelling and forecasting inflation has focused on domestic drivers, yet growing globalisation has fundamentally altered inflation dynamics. As Draghi (2015) remarked, 'Over the last decade, there has been a growing interest in the concept of global inflation. This is the notion that, in a globalised world, inflation is becoming less responsive to domestic conditions, and is instead increasingly determined by global factors'; see also Auer et al. (2019). Ignoring these global influences risks overstating the importance of domestic conditions and misleading macroeconomic policy. However, research explicitly examining how international factors shape the distribution of domestic inflation risks remains scarce.[1]

The interest in modelling and forecasting inflation risks is without discussion. Low and stable inflation is

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The numerical results presented in this manuscript were reproduced by the Editor-in-Chief on 13 December 2025.

[1] Research on worldwide fluctuations of average inflation has grown significantly over the last two decades after the seminal work of Rogoff (2003); see Forbes (2020) and Ha, Kose, Ohnsorge, and Unsal (2019) for descriptions of this research. The generalised consensus is about inflation being largely a global phenomenon; see among others Mumtaz and Musso (2021), Arango-Castillo et al. (2023), Ha, Kose, Ohnsorge, and Unsal (2019), Liu and Lee (2021), Carriero et al. (2022), Ha et al. (2023), Cepni and Clements (2024), and Medeiros et al. (2024) for some recent references.

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usually associated with economic growth and employment, while both very low and very high inflation are macroeconomic challenges; see for example Kilian and Manganelli (2008), who introduce a model in which the policymaker acts as a risk manager trying to balance the risks of inflation and output stability. Furthermore, due to recent geopolitical events such as the Russian invasion of Ukraine in February 2022, inflation underwent a synchronised increase above inflation targets in many advanced economies and, consequently, it has moved to the forefront of the public debate. Forecasting risks surrounding price stability in fact has implications for policymakers and central banks determining monetary policy, for businesses deciding how much to invest, and for households planning future savings and expenditures. In particular, policymakers need to assess the risk of having either very low or very high inflation; see the very recent paper by De Polis et al. (2025), who propose a new strategy, risk-adjusted inflation targeting (RAIT), which uses real-time assessments of inflation risks to guide central bank actions and communication, and dynamically adjusts monetary policy stance to counter their evolving balance.[2]

Within this context, in this paper, we forecast upside/downside risks (densities) of headline domestic inflation for a large set of $N = 115$ economies using as predictors underlying international factors.[3] The inflations considered, obtained from the very rich database of Ha et al. (2023), are observed monthly from January 1999 to December 2022 for countries that cover not only advanced (ADV) but also middle- and high-income emerging and developing (MHI-EMD) and low-income emerging (LI-EMD) economies. To estimate and forecast the density of inflation in each country, we use state-of-the-art econometric models based on factor-augmented quantile regression (FA-QR) as proposed by Bai and Ng (2008) and Ando and Tsay (2011) and popularised by Adrian et al. (2019) in the context of estimating growth at risk. Quantile regression allows for different impacts of international factors on different quantiles of the distribution of domestic inflation and, consequently, for potential asymmetries in the downside and upside inflation risks. Furthermore, among the predictors of the FA-QRs, we consider not only global international factors, but also group-specific international factors linked to countries in the same geographical regions, as in Medeiros et al. (2024), or to countries classified as either ADV, MHI-EMD, or LI-EMD, as in Ha, Kose, Ohnsorge, and Unsal (2019), Jorda and Nechio (2020), and Ha et al. (2023). To extract these factors, we use the multi-level dynamic factor model (ML-DFM) proposed by Breitung and Eickmeier (2016) for non-overlapping factors and extended by Rodríguez-Caballero and Caporin (2019) to models with overlapping factors such as that considered in this paper.

The exploration of the out-of-sample forecast loss functions carried out in this paper enables us to assess when international factors are relevant to inflation upside/ downside risks. We find that the impact of international factors is usually stronger in the tails than in the centre of the domestic inflation distribution. Furthermore, this impact is generally largest in the right quantiles and, consequently, on the risk of high inflation. We also find interesting heterogeneous patterns across countries. In concordance with the related literature, the impact of international factors is often stronger in ADV economies than in MHI-EMD and LI-EMD economies. With exceptions, global international factors matter for forecasting inflation at the upper quantiles in advanced European countries, being less relevant elsewhere. We also analyse the potential drivers of these results by fitting a panel model to the differences in the out-of-sample forecast loss functions of the FA-QRs and the quantile regressions estimated without international factors, as a function of trade integration, geopolitical risk, inflation level, economic growth, unemployment, and central bank independence; see Gali (2010) for a theory-based discussion of the global drivers of inflation, and Rogoff (2003), Ha, Kose, Ohnsorge, and Unsal (2019), Ha, Kose, Ohnsorge, and Yilmazkuday (2019), Jorda and Nechio (2020), Forbes (2020), and Cascaldi-Garcia et al. (2024) for detailed descriptions and references.[4] We show that, regardless of the quantile, the predictive power of international factors is larger the higher that inflation is. More trade openness also implies larger predictive power of international factors but only on those loss functions related to the right tail of the inflation distribution, while unemployment and per capita growth are relevant for the left tail. Geopolitical risk explains the out-of-sample performance of international factors in the median and right tail. Finally, international factors have larger predictive power for the right quantiles of inflation in countries with larger growth, while its predictive power is smaller in the left quantiles. Also, the predictive power of international factors is smaller in countries with larger central bank independence.

Our work is closely related to several very recent contributions in the literature. On the one hand, FA-QRs have been implemented before to estimate the quantiles of inflation, although using domestic instead of international factors; see for example Banerjee et al. (2024) and López-Salido and Loria (2024), who assume that inflation quantiles in a large number of economies depend on a set of domestic macroeconomic and financial variables, and Tabliabracci (2020), who explains euro area inflation as a function of a European business-cycle factor. On the other hand, several authors extract international factors from ML-DFMs to explain average inflation without analysing their impact on inflation risks; see for example Beck et al. (2015), Ha, Kose, Ohnsorge, and Unsal (2019), Ha et al. (2023), Cascaldi-Garcia et al. (2024), Kim and Kwark (2024), and Medeiros et al. (2024).[5] However,

[2] See also Mankiw (2024), who recommends that central bankers offer as target for inflation a range instead of a single figure, recognising how imprecise their ability to control inflation is.

[3] Headline inflation is still the long-run target of most central banks around the world. For example, the long-run inflation objective of the Federal Open Market Committee is defined in terms of the headline price index. Similarly, the European Central Bank considers headline inflation to be the appropriate measure for assessing the achievement of the price stability objective.

[4] Due to data availability, this analysis is carried out using annual data for a subsample of 38 countries.

works merging international factors and inflation risks are very scarce. Busetti et al. (2021) propose using expectile regression instead of quantile regression to model the distribution of core inflation as a function of domestic and foreign output gap, focusing their analysis on the euro area. More closely, Nazlioglu et al. (2025) use the same database considered in this paper to obtain forecasts of densities of headline inflation based on global international factors. There are four main differences between our approach and that of Nazlioglu et al. (2025). First, it is important to note that they extract international factors using the quantile factor model of Chen et al. (2021). Different factors are extracted from different empirical quantiles of the data, with the potential issue of having too few observations to estimate factors in extreme quantiles of the distribution. Second, they use annual data instead of monthly data, with the problem associated with the small number of observations used to estimate extreme quantiles being even more relevant. In fact, in the rolling-window out-of-sample forecasting exercise, their factors are extracted using just 23 annual observations. Third, they do not control for the stationarity of inflation and consider a mixture of I(1) and I(2) variables. Finally, they only extract global international factors, without considering the possibility of specific factors common to particular regions and/or levels of development.

The rest of the paper is organised as follows. Section 2 describes the main empirical stylised facts of the inflation data analysed in this paper and reports some results of the in-sample models used to forecast the densities of inflation. Section 3 presents the main results of the out-of-sample forecasting assessment of international factors. In Section 4, we report the results of fitting a panel model to investigate the variables behind the forecast ability of international factors. Finally, Section 5 summarises the main takeaways. This paper also contains several online appendices. The description of the methodology used to estimate the domestic inflation densities based on FA-QR models with the factors extracted using ML-DFMs is relegated to Appendix A, while Appendix B reports further empirical results.

2. Data and in-sample analysis

In this section, we describe the data and report some results of the in-sample conditional quantiles of inflation estimated in each country.

2.1. Data

We obtain monthly observations of headline CPI from January 1999 to December 2022 ($T = 288$) for a set of $N = 115$ countries around the world from the novel Global Database on Inflation (GDI) constructed by Ha et al. (2023). The GDI compiles data on prices from multiple sources, including various international institutions as well as a large number of country-specific sources; see Pitarakis (2023), Medeiros et al. (2024), and Nazlioglu et al. (2025), who also consider the GDI.[6] Table 1 lists all countries considered in this paper, which are classified according to two different criteria. First, according to the International Monetary Fund (IMF), they are classified into 35 ADV economies and 80 emerging and developing economies. Furthermore, within this last group, according to the World Bank Group country classifications by income level for FY24 (July 1, 2023, to June 30, 2024), we consider the group of 42 MHI-EMD economies and those of 38 LI-EMD economies.[7] The top panel of Fig. 1 represents the world map with the distribution of countries classified according to their income. Second, the countries in our database are classified according to their geographical area as follows: Asia and Pacific (21), Europe and Central Asia (40), America (21), and the Middle East and Africa (33). The bottom panel of Fig. 1 represents the world map with the distribution of countries classified according to their geographical region. Note that there is an important overlap between countries classified according to their geographical region and according to their development level.

For each country considered, CPI prices are transformed to annualised month-on-month (*mom*) inflation with each series of inflation sequentially cleaned of seasonal effects and outliers.[8] To deseasonalise, we follow Harvey (2006) and fit a univariate structural time series model to each series of inflation with the seasonal component estimated using the Kalman filter and subtracted from the original observations. After deseasonalising each series, we follow McCracken and Ng (2016) and identify outliers as observations that deviate from the sample median by more than 10 interquartile ranges (IRs), substituting them by $10 \times IR$ multiplied by the sign; see Online Appendix B for a list of the outliers.

Fig. 2 plots the corrected series of inflation separately in the ADV, MHI-EMD, and LI-EMD economies. It is remarkable that with the exception of the ADV economies, we observe episodes of what can be called hyperinflation in a large proportion of economies. However, in the last few years since the Russian invasion of Ukraine, ADV

[5] Ciccarelli and Mojon (2010) and Cepni and Clements (2024) also propose extracting the common international factors of inflation in OECD countries by principal components and in emerging European countries by partial least squares, respectively, and use the factors to estimate factor-augmented predictive regressions for average inflation.

[6] Although headline inflation data in the GDI are available before 1999 for 209 countries, we select $N = 115$ countries in order to have a balanced dataset since 1999, which corresponds with the introduction of the euro and the pegging of European currencies. Furthermore, several countries, namely Belarus, Haiti, Sudan, Malawi, Tanzania, the United Republic of Gambia, the Maldives, and Hong Kong SAR (China), are excluded from our analysis due to the very atypical behaviour of their inflation series.

[7] The World Bank Group classifications are updated each year on July 1, based on the GNI per capita of the previous calendar year. Source: https://blogs.worldbank.org/en/opendata/new-world-bank-group-country-classifications-income-level-fy24.

[8] By considering *mom* instead of year-over-year (*yoy*) inflation, we avoid issues associated with potential non-stationarity; see Stock and Watson (1999), who found little difference in $I(1)$ and $I(2)$ factor model forecasts for prices. Furthermore, cleaning the seasonal effects and outliers is standard practice in this literature; see Camacho et al. (2015), who conclude that factor extraction is not affected by deseasonalisation.

Table 1
List of countries classified by income together with their codes and geographical regions.

Developed			Middle-high emerging			Low emerging		
Country	Code	Region	Country	Code	Region	Country	Code	Region
Austria	AUT	Europe	Albania	ALB	Europe	Algeria	DZA	Africa
Belgium	BEL	Europe	Antigua and Barbuda	ATG	America	Bangladesh	BGD	Asia
Canada	CAN	America	Armenia	ARM	Europe	Benin	BEN	Africa
Croatia	HRV	Europe	Azerbaijan	AZE	Europe	Bolivia	BOL	America
Cyprus	CYP	Europe	Bahamas	BHS	America	Burkina Faso	BFA	Africa
Czech Republic	CZE	Europe	Botswana	BWA	Africa	Burundi	BDI	Africa
Denmark	DNK	Europe	Brazil	BRA	America	Cambodia	KHM	Asia
Estonia	EST	Europe	Bulgaria	BGR	Europe	Cameroon	CMR	Africa
Finland	FIN	Europe	Chile	CHL	America	Central African Republic	CAF	Africa
France	FRA	Europe	China	CHN	Asia	Chad	TCD	Africa
Germany	DEU	Europe	Colombia	COL	America	Congo, Rep.	COG	Africa
Greece	GRC	Europe	Costa Rica	CRI	America	Cote d'Ivoire	CIV	Africa
Iceland	ISL	Europe	Dominican Republic	DOM	America	Egypt, Arab Rep.	EGY	Africa
Ireland	IRL	Europe	Ecuador	ECU	America	Ghana	GHA	Africa
Israel	ISR	Africa	El Salvador	SLV	America	Guinea-Bissau	GNB	Africa
Italy	ITA	Europe	Equatorial Guinea	GNQ	Africa	Honduras	HND	America
Japan	JPN	Asia	Fiji	FJI	Asia	India	IND	Asia
Korea, Rep.	KOR	Asia	Gabon	GAB	Africa	Jordan	JOR	Africa
Latvia	LVA	Europe	Georgia	GEO	Europe	Kenya	KEN	Africa
Lithuania	LTU	Europe	Guatemala	GTM	America	Lao, PDR	LAO	Asia
Luxembourg	LUX	Europe	Hungary	HUN	Europe	Madagascar	MDG	Africa
Macao SAR, China	MAC	Asia	Indonesia	IDN	Asia	Mauritania	MRT	Africa
Malta	MLT	Africa	Jamaica	JAM	America	Mongolia	MNG	Asia
Netherlands	NLD	Europe	Kazakhstan	KAZ	Asia	Morocco	MAR	Africa
Norway	NOR	Europe	Malaysia	MYS	Asia	Niger	NER	Africa
Portugal	PRT	Europe	Mauritius	MUS	Africa	Pakistan	PAK	Asia
Singapore	SGP	Asia	Mexico	MEX	America	Philippines	PHL	Asia
Slovakia	SVK	Europe	Moldova, Rep.	MDA	Europe	Rwanda	RWA	Africa
Slovenia	SVN	Europe	North Macedonia	MKD	Europe	Samoa	WSM	Asia
Spain	ESP	Europe	Paraguay	PRY	America	Sao Tome and Principe	STP	Africa
Sweden	SWE	Europe	Peru	PER	America	Senegal	SEN	Africa
Switzerland	CHE	Europe	Poland	POL	Europe	Solomon Islands	SLB	Asia
Taiwan, China	TWN	Asia	Romania	ROU	Europe	Togo	TGO	Africa
United Kingdom	GBR	Europe	Saudi Arabia	SAU	Africa	Tunisia	TUN	Africa
United States	USA	America	Serbia	SRB	Europe	Uganda	UGA	Africa
			South Africa	ZAF	Africa	Ukraine	UKR	Europe
			St. Kitts and Nevis	KNA	America	Vietnam	VNM	Asia
			St. Lucia	LCA	America	Zambia	ZMB	Africa
			Thailand	THA	Asia			
			Tonga	TON	Asia			
			Turkey	TUR	Europe			
			Uruguay	URY	America			

economies are amid their worst inflation pressure over the sample period, due to a plethora of unprecedented developments. In some ADV countries with a long history of low and stable inflation, monthly inflation has soared to 40%, also showing episodes of −30% deflation. Furthermore, the extent and pace of this surge are not just a recurring problem in ADV economies. It is becoming an entrenched phenomenon across the world. These unprecedented extreme movements in inflation could be attributed to, among other causes, the strong rebound in aggregate demand caused by the extraordinary policy response to the Covid-19 pandemic and to global supply constraints and shock waves through international commodity markets triggered by the Russian invasion of Ukraine; see for example Gordon and Clark (2023).

2.2. In-sample estimation of inflation densities

The following FA-QR is estimated for each country and quantile for $\tau = 0.05, 0.25, 0.50, 0.75, 0.95$:

$$q_\tau\left(y_{it+1}|y_{it}, F_t\right) = \mu(\tau, i) + \phi(\tau, i)y_{it} + \sum_{k=1}^{r}\beta_k(\tau, i)F_{kt}, \quad (1)$$

where y_{it} is the inflation observed in country i at time t, for $i = 1, \ldots, N$, and $t = 1, \ldots, T$. The factors are extracted from an ML-DFM fitted to $Y_t = (y_{1t}, \ldots, y_{Nt})'$; see Online Appendix A.1 for details on the estimation methodology and Online Appendix B for a detailed description of the extracted factors used to estimate model (1). Note that, as it is often done in this literature, we adopt a 'direct' scheme. Furthermore, it is important to remark that the international factors used to estimate the FA-QR are global factors loading on all inflations in the sample and specific factors common to inflations in countries in a given geographical region or to countries grouped according to their income level. A big-picture overview of the estimation results for all $N = 115$ inflations and quantiles is given in Fig. 3, which represents the combinations of countries and quantiles for which the F-test for the joint significance of the factors is rejected; see also Online Appendix B for a detailed description of the estimation results. We can observe that the international

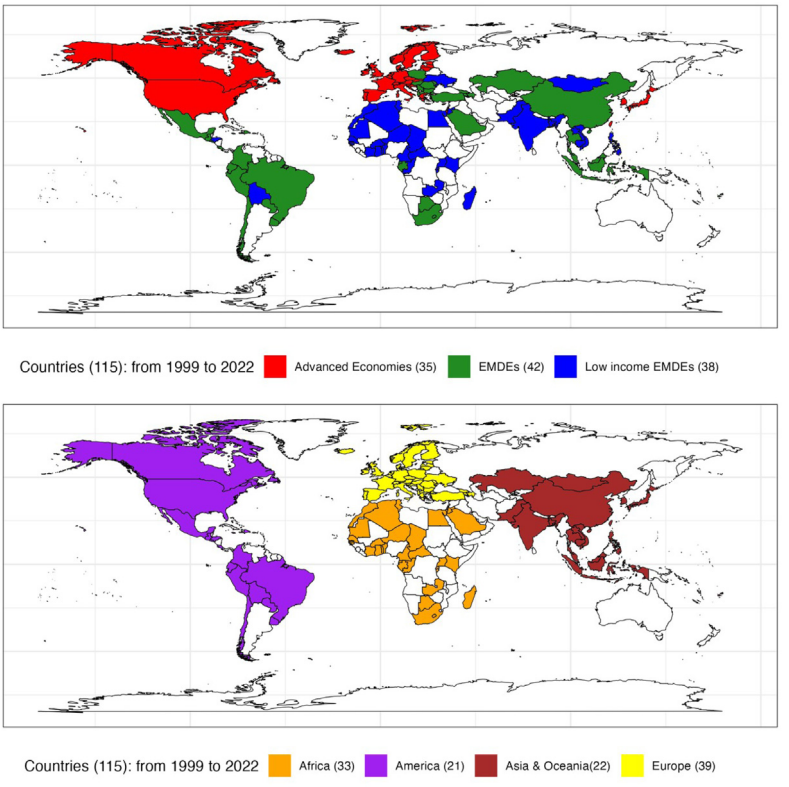


Fig. 1. Maps of countries classified according to income (top) and region (bottom).

factors are jointly significant for at least one quantile in all economies in Europe and Asia. This is also the case in America, but for Brazil and the Dominican Republic, and in Africa, but for Zambia, Benin, and Egypt. When looking at the results for different quantiles, we can observe that the international factors are not significant in a large proportion of economies when $\tau = 0.5$. However, when τ is either 0.05 or 0.95, the international factors are significant in a large number of economies. All in all, we find that international factors are important when modelling the distribution of inflation all around the world.

For each country, in-sample one-step-ahead conditional densities of inflation, $\tilde{k}_{it+1}$, are obtained by using the estimated quantiles proposed by Azzalini and Capitanio (2003). Fig. 4, which plots them for China, Germany, Mexico, and the US, shows that the Russian invasion has a very strong effect on Germany, while inflation densities are very stable in China.

Finally, based on $\tilde{k}_{it+1}$, we obtain one-step-ahead in-sample quantile-related measures of upside and downside risks to inflation. In particular, we estimate the inflation at risk (IaR) and deflation at risk (DaR). The IaR is the probability that inflation is larger than a given threshold π^*, and, therefore, measures the probability of high-inflation. The IaR of country i at time $t+1$ for threshold π^* is given by

$$IaR_{it+1}(\pi) = \int_{\pi^*}^{\infty} \tilde{k}(y_{it+1})dy_{it+1}. \quad (2)$$

Alternatively, we can measure the probability of deflation using the DaR, which is defined as the probability that inflation falls below a certain threshold, π^{**}. The DaR is given as follows[9]:

$$DaR_{it+1}(\pi) = \int_{-\infty}^{\pi^{**}} \tilde{k}(y_{it+1})dy_{it+1}. \quad (3)$$

In this paper, we consider measuring the DaR with $\pi^{**} = 0$ and the IaR with three different threshold levels, namely, $\pi^* = 1\%$, 2%, and 3%; see Tabliabracci (2020), who proposes $\pi^{**} = 0$ and $\pi^* = 2\%$, Garrat and Petrella (2022),

[9] Kilian and Manganelli (2007) propose an alternative way of measuring the risk of inflation linked to the concept of expected loss in the context of financial risk. They define the deflation risk and the risk of excessive inflation as probability-weighted functions of the deviations of inflation from the threshold π. These measures include the IaR and DaR as special cases. Furthermore, they propose a balance of risk measures based on the deflation risk and the risk of excessive inflation. They implement these measures to estimate the risk of inflation based on an AR-GARCH model using a bootstrap to obtain the density; see Garrat and Petrella (2022) for an application in which inflation risks depend on commodity prices, and see Wang et al. (2024) for an application using Bayesian procedures to estimate the inflation density depending on selected indicators and with time-varying parameters modelled as random walks. Alternatively, Banerjee et al. (2024) propose measuring inflation risk using entropy measures.

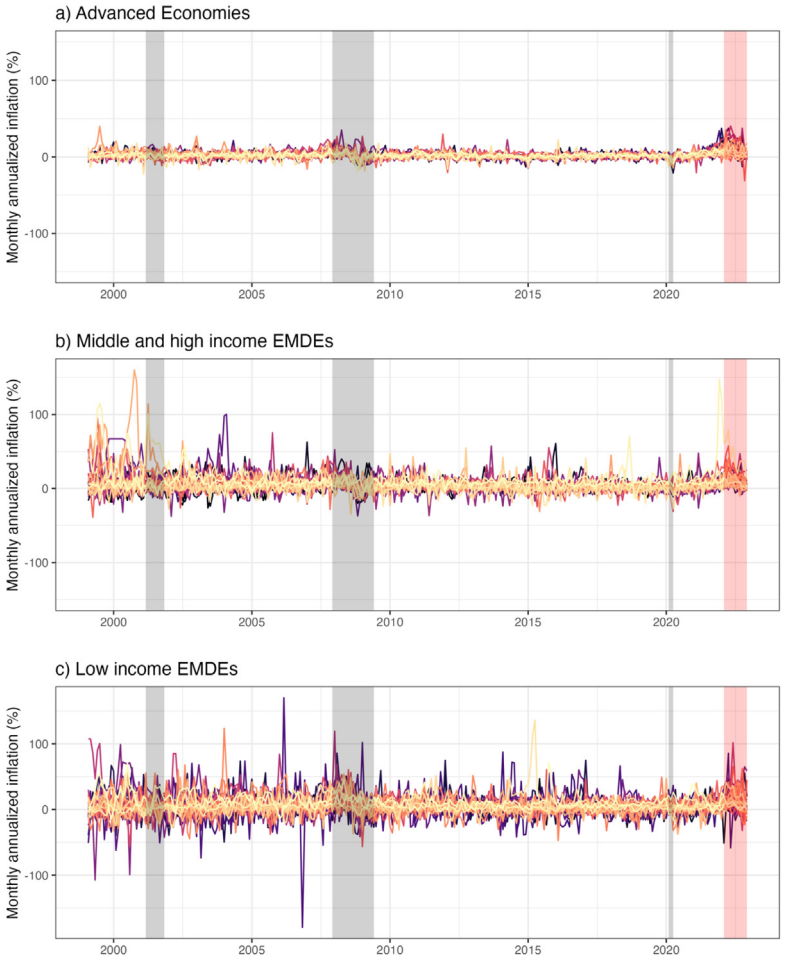


Fig. 2. Deseasonalised annualised monthly inflations grouped by development level. The grey shaded areas represent recession periods as identified by the US National Bureau of Economic Research (NBER), and the red shaded areas represent the Russo–Ukrainian war period. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

who consider $\pi^* = 3\%$ for inflation in the US, and Balatti (2020), who report the range of inflation targets for a large number of central banks.[10] Fig. 5 plots these estimates for all $N = 115$ economies, highlighting those corresponding to the G7 and G5 economies. It is remarkable that while the Russian invasion increases the probability of high inflation (over 3%) all over the world, the effect in Europe is strongest, moving from probabilities between 1% and 40% to probabilities over 90% that have only been observed after the global financial crisis. On the other hand, in some countries such as India, there is no impact from the Russian invasion. Fig. 5 also shows that the probability of deflation is usually low, with a peak after the Covid-19 crisis.

3. Out-of-sample forecasts of inflation densities

In this section, we assess the forecasting content of international global and group-specific factors for real-time out-of-sample one-step-ahead forecasts of the quantiles of the conditional density of domestic inflation for each economy in the sample.

3.1. Obtaining out-of-sample forecasts

The sample size is divided into an estimation period from February 1999 to December 2011 ($R = 155$ observations) and an out-of-sample period from January 2012 to December 2022 ($P = 132$ observations), with $T = R + P$; see Hansen and Timmermann (2012) for rules to choose R and P. Real-time one-step-ahead forecasts of a particular quantile of the conditional domestic density of inflation in each economy are obtained using the estimated FA-QR model in (1) with the model parameters estimated using a rolling-window scheme, as follows:

$$\hat{q}_\tau(y_{it+1}|y_{it}, F_t) = \hat{\theta}_t X_{it}, \tag{4}$$

where, for $t = R, \ldots, T$, $\hat{\theta}_t$ denotes estimates of the parameters of the FA-QR model obtained with observations from $t - R + 1$ up to time t; and $X_{it} = \left(y_{it}, \hat{F}_{gt}, \hat{F}_{st}\right)'$, with

[10] In practice, the European Central Bank considers that price stability is best maintained by aiming for 2% inflation over the medium term.

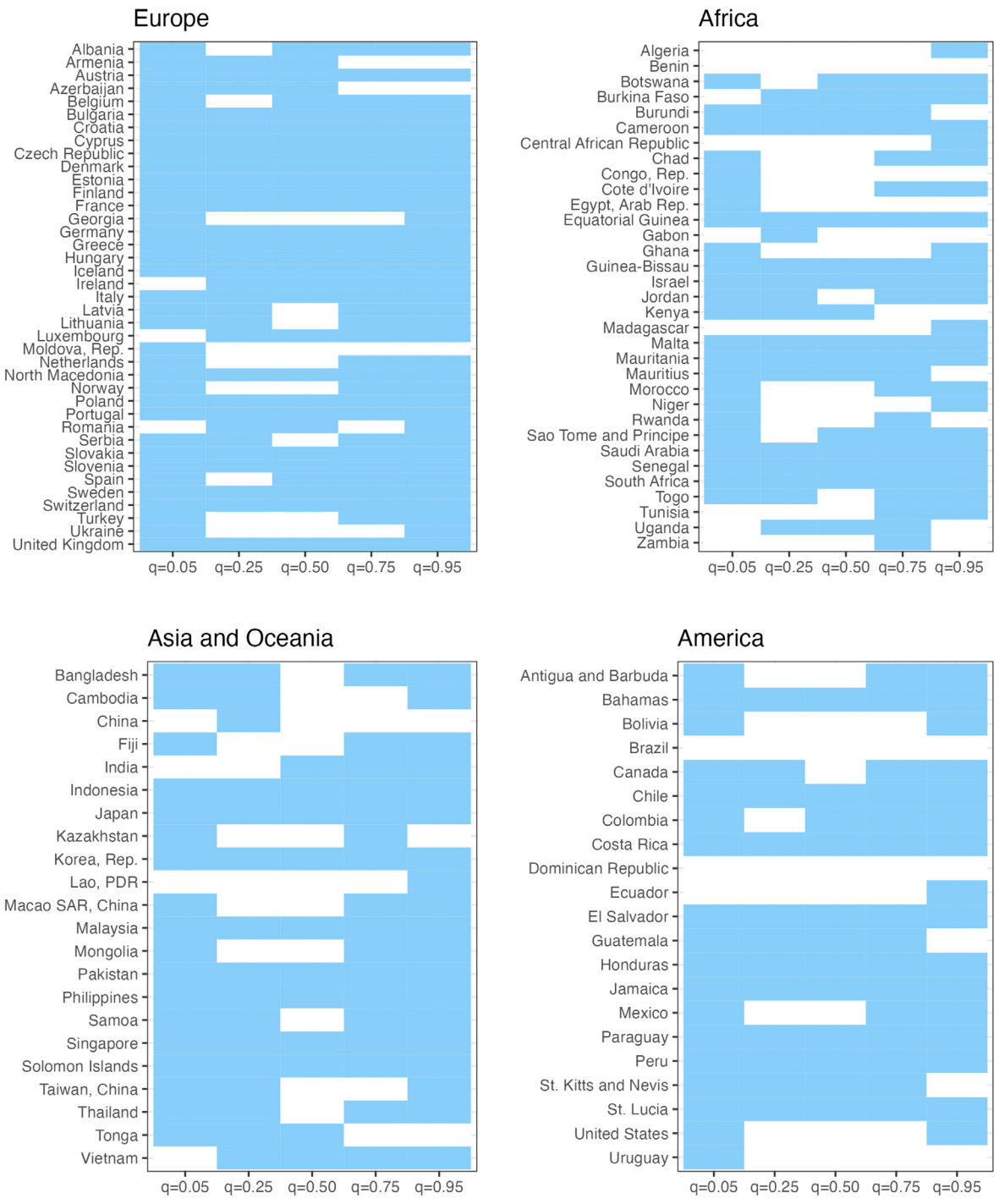


Fig. 3. Tests of joint significance of the parameters of the FA-QR models associated with international factors. In blue: rejection of the null when p-value is smaller than 0.10. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the factors extracted from the ML-DFM in each rolling sample; see Amburgey and McCracken (2022) for the relevance of real-time factor extraction for assessing the forecast performance of FA-QR models. The first rolling-window vintage is based on observations from February 1999 to December 2011, while the last one is based on observations from February 2010 to December 2022. Fig. 6, which plots the extracted factors obtained in each rolling-window vintage, shows that they are rather consistent through vintages.

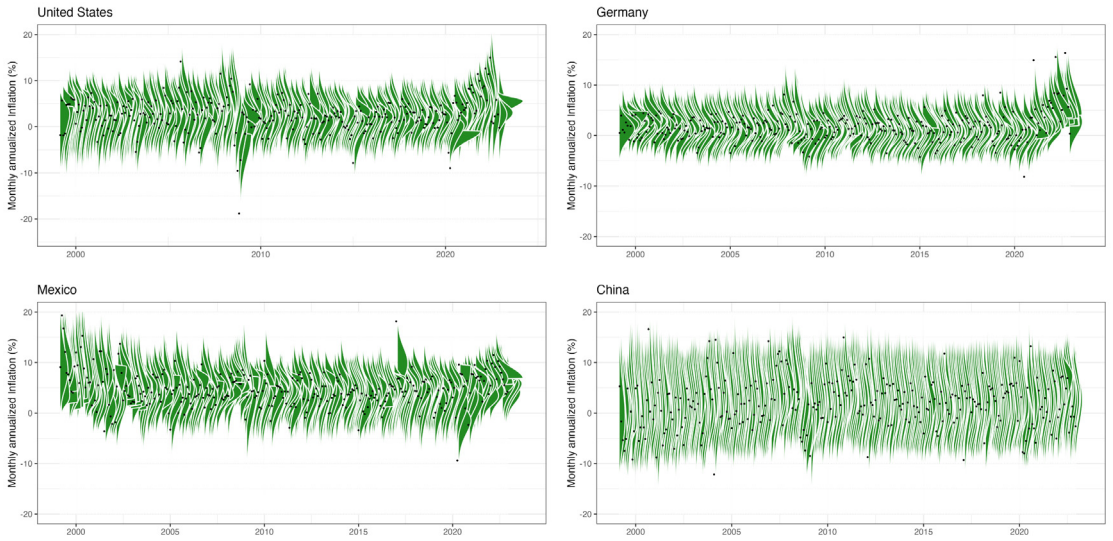


Fig. 4. In-sample one-step-ahead predictions of domestic inflation densities in the US, Germany, Mexico, and China obtained using the FA-QR model. The bullets represent the observed inflation.

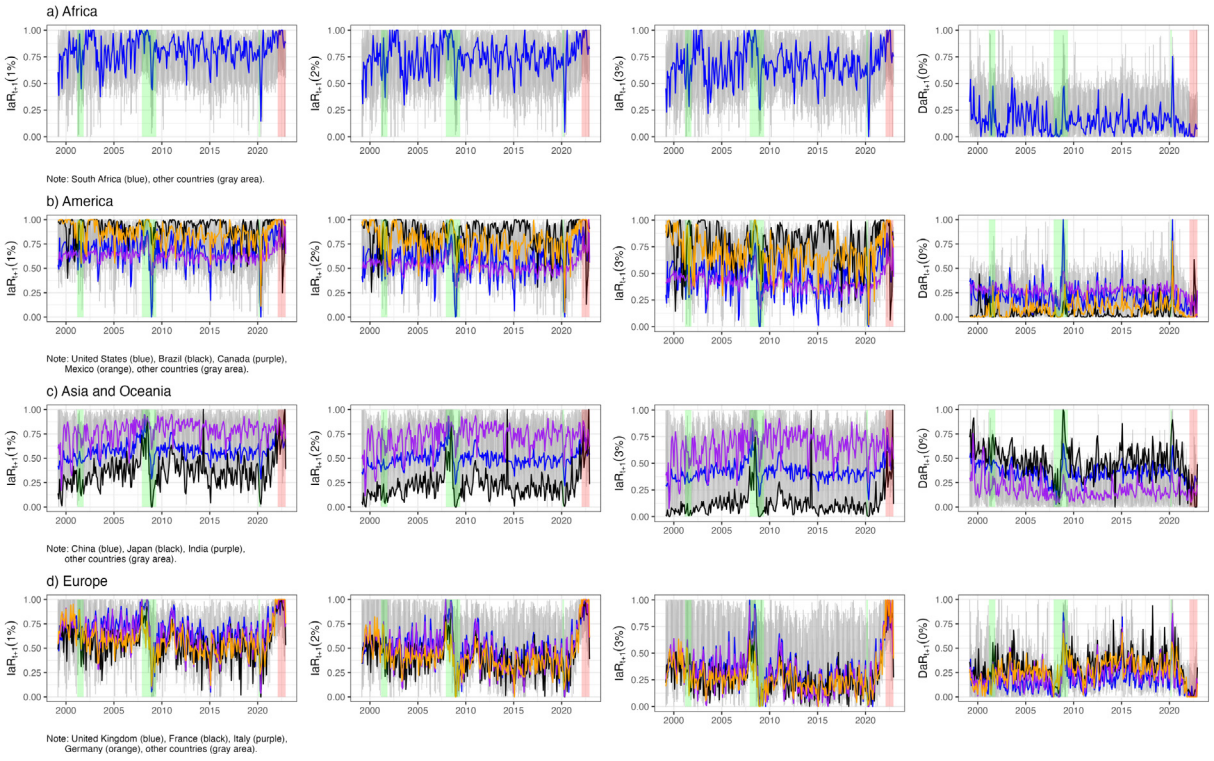
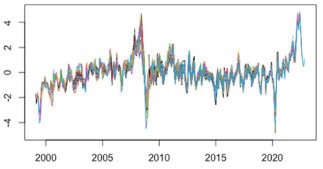
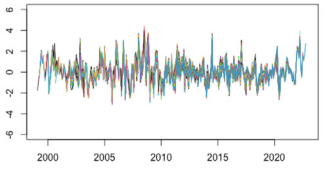


Fig. 5. Estimates of $IaR_{it+1}(1)$ (first column), $IaR_{t+1}(2)$ (second column), $IaR(3)_{t+1}$ (third column), and $DaR_{it+1}(0)$ (fourth column), for all economies in Africa (first row), America (second row), Asia (third row), and Europe (fourth row).

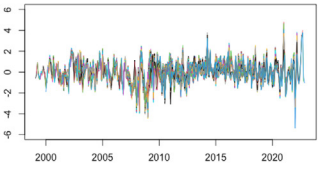
Note that using a rolling-window scheme attenuates potential issues associated with parameter instabilities during the forecasting period; see the discussions by Giacomini and Rossi (2010) and Inoue et al. (2017). We consider the possibility of time-varying parameters with the FA-QR models by analysing the rolling-window estimates of the parameters of the FA-QRs. The evolution of the estimated parameters of the international factors is illustrated for the US in Fig. 7, which plots them together with their 95% confidence bounds for $\tau =$



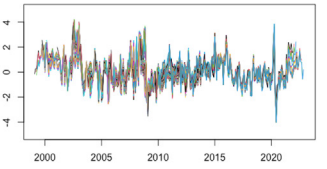
(a) Global factor



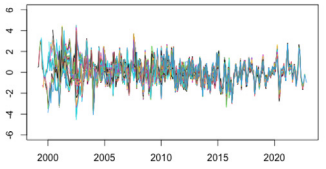
(b) African factor



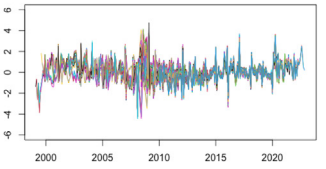
(c) Advanced economies factor



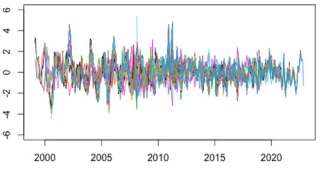
(d) American factor



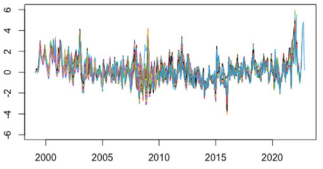
(e) HMI economies factor



(f) Asian factor



(g) LI economies factor



(h) European factor

Fig. 6. Vintages of factors extracted in each estimation period in the rolling window.

0.05, 0.25, 0.50, 0.75, and 0.95.[11] We can observe that all parameters are mostly constant for all quantiles except when looking at the 95% quantile of the global and region-specific factors. It is remarkable that the parameters associated with these factors are increasing through time, but for the very last years associated with the invasion of Ukraine; see Forbes (2020), Kim and Kwark (2024), Shin and Kang (2013), and Elsayed et al. (2021) for the relative importance of the global factor changing over time.

We obtain out-of-sample one-step-ahead forecasts of the conditional densities of inflation $\tilde{k}_{it+1}$. For illustrative purposes, Fig. 8 plots these forecasts in the US, Germany, Mexico, and China, which are very similar to the in-sample predictions plotted in Fig. 4. We can observe that with the Covid-19 pandemic, in 2020, the forecasts of inflation densities in the US and Mexico are rather erratic. However, in Germany, the forecasts are less affected by the pandemic, whereas both the average and dispersion of the inflation densities are clearly larger after the invasion of Ukraine. Finally, note that the impact of the pandemic and the invasion on the inflation densities in China is

[11] Estimates for other economies are available upon request.

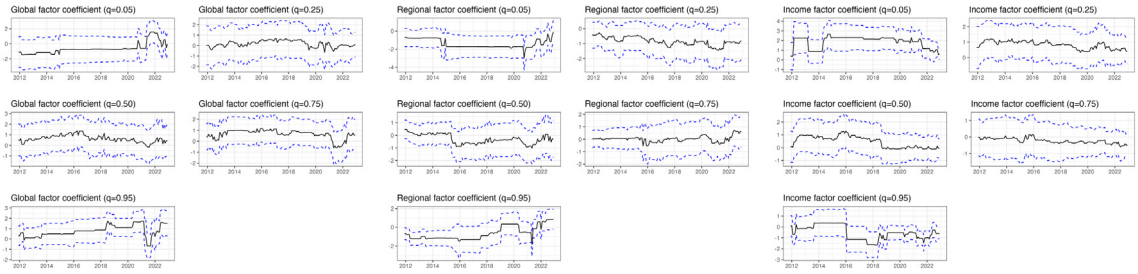


Fig. 7. Estimated parameters of the international factors of the FA-QR models for the US and $\tau = 0.05, 0.25, 0.50, 0.75$, and 0.95, obtained in the rolling window: global factor (first column), regional factor (second column), and economic development factor (third column).

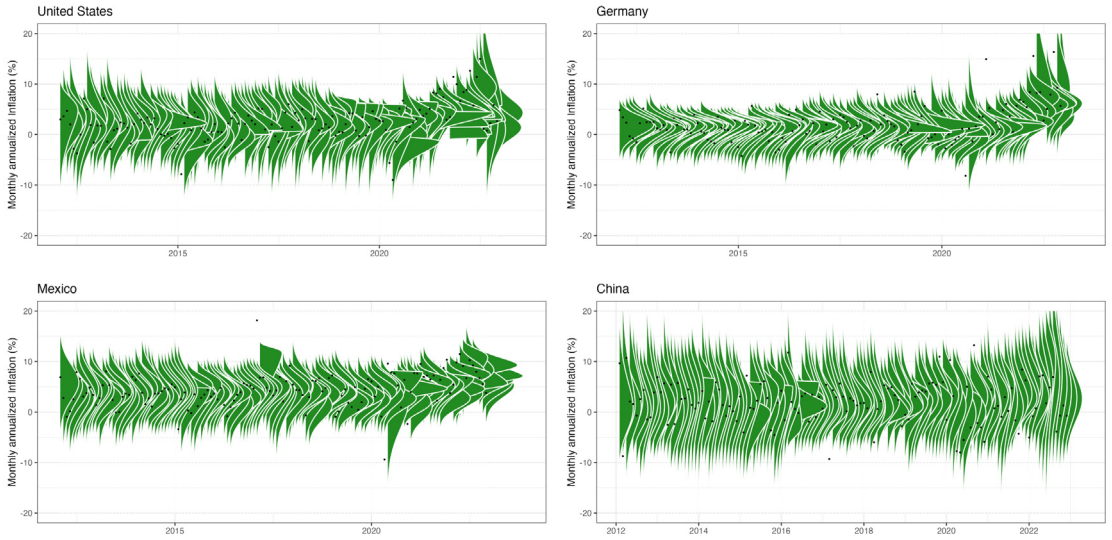


Fig. 8. Out-of-sample one-step-ahead forecasts of domestic inflation densities in the US, Germany, Mexico, and China obtained using the FA-QR model. The bullets represent the observed inflation.

very mild as a consequence of China not being affected by international factors.

Finally, Fig. 9 plots out-of-sample one-step-ahead forecasts of the IaR for thresholds of 1%, 2%, and 3% and of the DaR for a threshold of 0%. The results are rather different when looking at countries in different continents. In South Africa, we can observe a very low probability of deflation, yet regardless of the threshold used, the probability of high inflation is similar and large. A similar result is observed in Brazil and somehow in India and Mexico, although in these latter two countries, the probability of high inflation is slightly smaller and the probability of deflation slightly larger. We can also observe that the probabilities in Canada and the US, which are very similar, imply similar levels of probability of very high inflation (over 3%) and deflation. A similar situation can be observed in China and the European countries. A completely different pattern is observed in Japan, with the probability of deflation being larger than that of high inflation.

3.2. Evaluating density forecasts

For each country, the one-step-ahead forecasts of the densities $\tilde{k}_{it+1}$ are evaluated using the test of the correct calibration of forecasts proposed by Rossi and Sekhposyan (2019), which relies on the probability integral transform; see for example Wang et al. (2024) and Lenza et al. (2025) for applications. The probability integral transform, originally proposed by Diebold et al. (1998) to evaluate forecasts of densities, is defined as the cumulative density function corresponding to $\tilde{k}_{it+1}$ evaluated at the true observation, y_{it+1}. The test of correct calibration forecasts, which is based on the Kolmogorov–Smirnov test, allows for a graphical representation. Fig. 10, which plots the results of the probability integral transform test of Rossi and Sekhposyan (2019) for the inflation density forecasts of the G7 and G5 economies, shows that the null hypothesis of the FA-QR density forecasts being well calibrated is not rejected. The FA-QR densities produce well-calibrated density forecasts for inflation in the G7 and G5 countries considered.

We also compare the relative performance of international factors to forecast inflation densities for each economy and quantile by using proper scoring rules; see Gneiting and Katzfuss (2014) and Kleen (2024) for excellent reviews of scoring rules. In particular, the out-of-sample accuracy of $\hat{q}_\tau(y_{it+1}|y_{it}, F_t)$ is assessed using the quantile score (QS), given by the following consistent

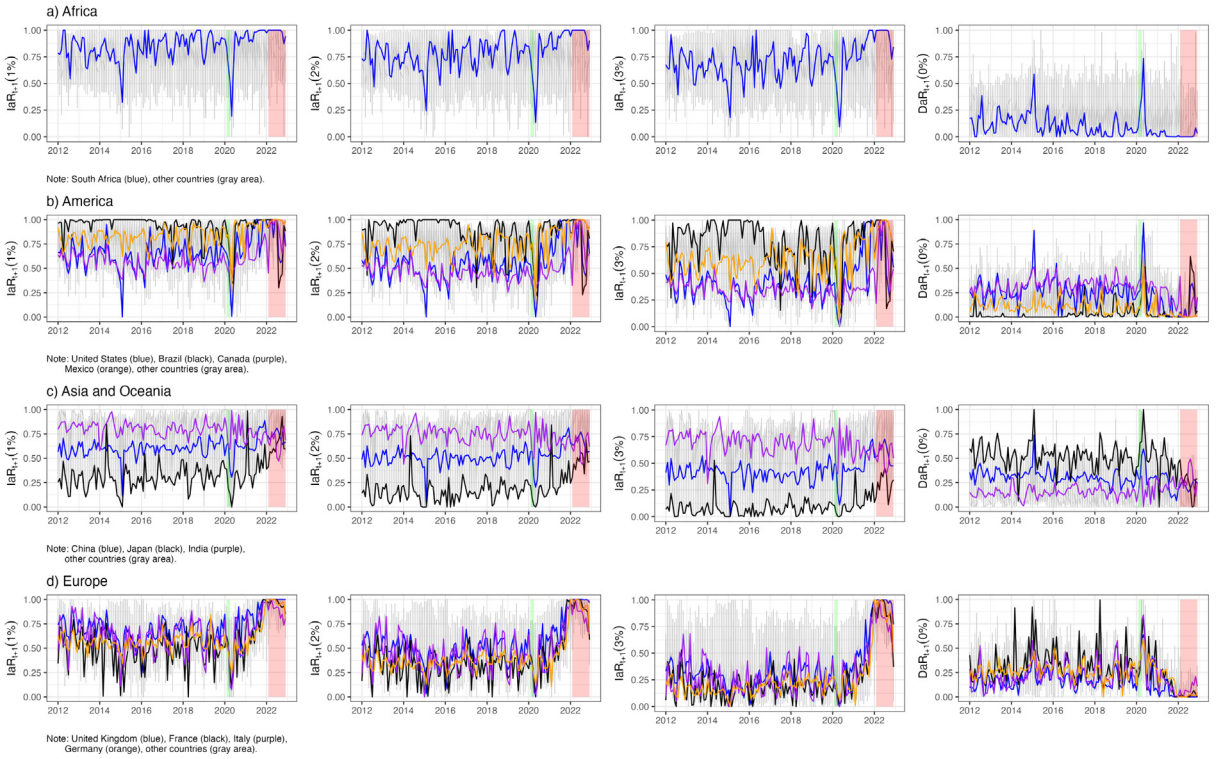


Fig. 9. One-step-ahead out-of-sample forecasts of $IaR_{it+1}(1)$ (first column), $IaR_{t+1}(2)$ (second column), $IaR(3)_{t+1}$ (third column), and $DaR_{it+1}(0)$ (fourth column) for all economies in Africa (first row), America (second row), Asia (third row), and Europe (fourth row).

asymmetric piecewise linear tick loss function:

$$QS(i,\tau) = \frac{1}{P}\sum_{t=R+1}^{T}\left(\mathbb{1}(y_{it} \le \hat{q}_\tau(y_{it})) - \tau\right)\left(\hat{q}_\tau(y_{it}) - y_{it}\right). \quad (5)$$

The lower the score, the better the probability forecast.

Alternatively, instead of focusing on just one quantile of the distribution, as in (5), one can use scoring rules specified in terms of the full density function. Among them, one of the most popular is the continuous ranked probability score (CRPS) proposed by Matheson and Winkler (1976) and popularised by Gneiting (2007). In particular, we consider the following weighted CRPS proposed by Gneiting and Ranjan (2011), in which the weighting emphasises particular regions of interest of the density:

$$CRPS(i) = \frac{1}{J}\sum_{j=1}^{J}\omega(\tau_j)QS(i,\tau_j), \quad (6)$$

where $J = 5$, $\tau_1 = 0.05, \ldots, \tau_5 = 0.95$, and $\omega(\tau)$ is one of the following weighting schemes: (i) $\omega(\tau) = (\tau - 1)^2$, denoted as the left tail (CRPS-L); (ii) $\omega(\tau) = \tau^2$, denoted as right tail (CRPS-R); and (iii) $\omega(\tau) = 1$, denoted as equal (CRPS-E). Recently, Kleen (2024) showed that the QS and CRPS are not only invariant to transformations in the data but also insensitive to the fact that the observed value of the target variable is not necessarily equal to the value of the true predictand. It is also important to mention that the CRPS is less sensitive to extreme events, which is particularly relevant when evaluating one-step-ahead inflation forecasts; see Gneiting (2007) and Panagiotelis and Smith (2008).[12]

We compute the scoring function $QS(i,\tau)$ for $i = 1, \ldots, 115$, and $\tau = 0.05, 0.25, 0.5, 0.75$ and 0.95, and the $CRPS(i)$ statistic, for each of the three weighting schemes: CRPS-L, CRPS-R, and CRPS-E. We also compute the corresponding ratios $\frac{QS^{FA-QR}}{QS^{AR-QR}}$ and $\frac{CRPS^{FA-QR}}{CRPS^{AR-QR}}$ with respect to the benchmark AR-QR model. Ratios smaller than one favour the FA-QR over the AR-QR forecasts. Next, we describe the main results.

Fig. 11 plots the ratios $\frac{QS^{FA-QR}}{QS^{AR-QR}}$ for $\tau = 0.05, 0.50,$ and 0.95, and $\frac{CRPS^{FA-QR}}{CRPS^{AR-QR}}$ for the CRPS-E, CRPS-R, and CRPS-L losses.[13] We can observe that regardless of the loss used, all ratios (except QS(0.05)) are below one in most European countries. This implies the very strong power of international factors to forecast the density of inflation in European economies. The results are not so clear when looking at all other continents, in which the analysis has to be done separately for each country.

We assess the forecast performance of international factors by testing for the equal predictive ability (based on either the QS or CRPS) of a given FA-QR model with respect to the quantile regression model in (1) without factors, i.e. with only a constant and the autoregressive

[12] Recently, Iacopini et al. (2022) proposed an asymmetric version of the weighted CRPS.

[13] The ratios corresponding to FA-QR models in which only one of the three factors is included are reported in Online Appendix B.

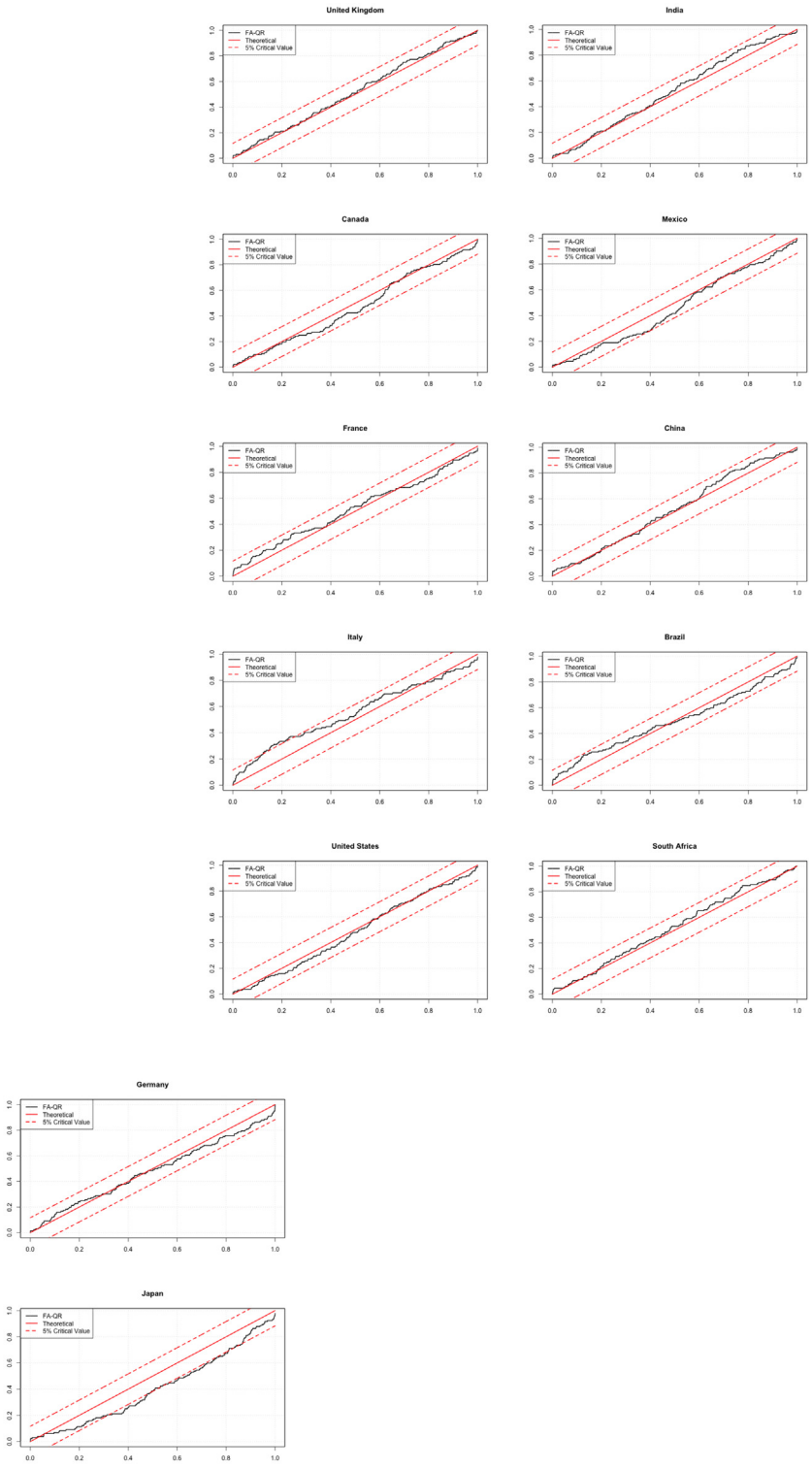


Fig. 10. Probability integral transforms of the FA-QR forecast densities: G7 countries (left column) and G5 countries (right column).

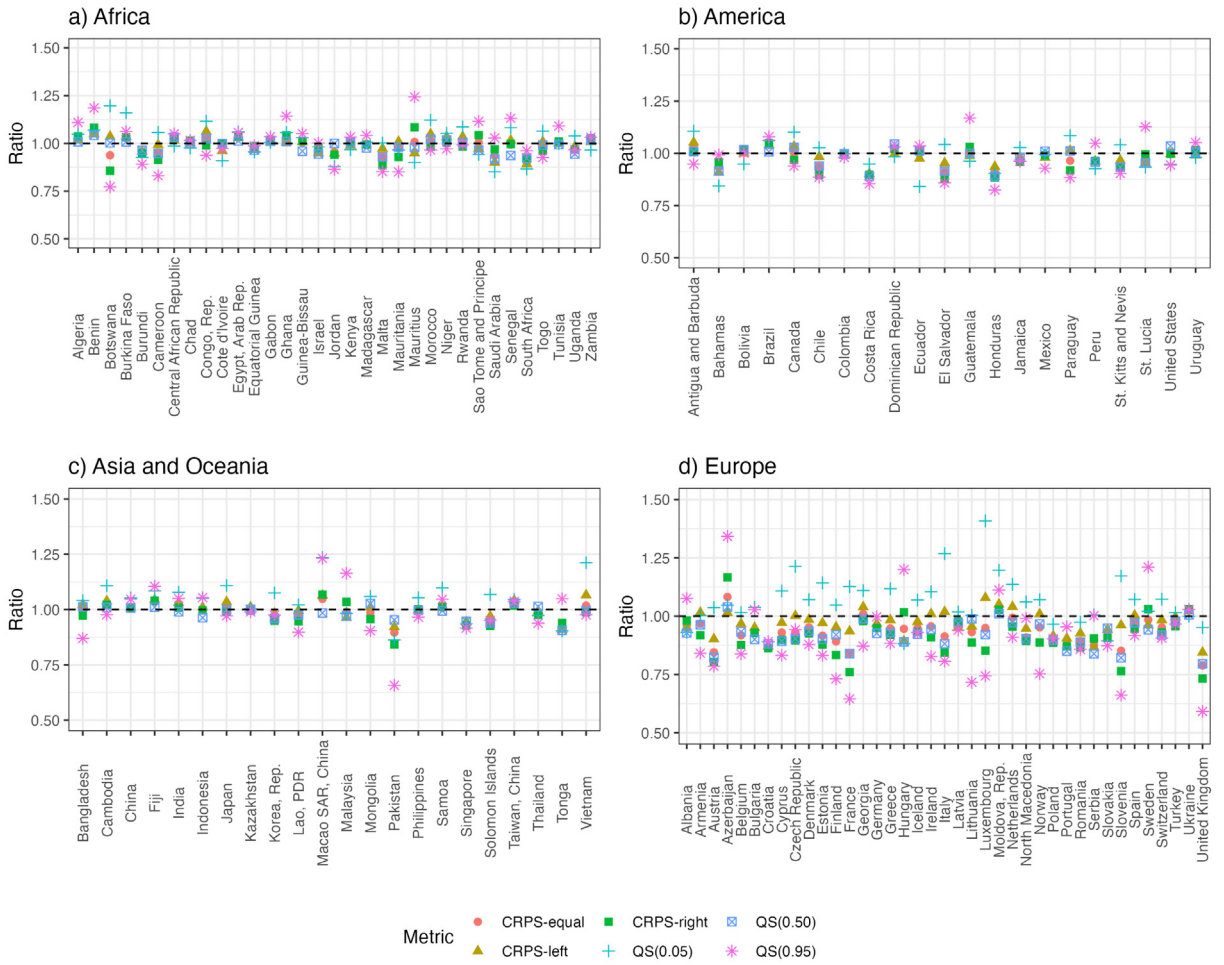


Fig. 11. QS ratios for $\tau = 0.05, 0.50$, and 0.95, and CRPS-E, CRPS-R, and CRPS-L losses between the FA-QR and AR-QR models.

term, denoted as the AR-QR model. The significance of the difference between losses of the FA-QR and AR-QR models is tested using the one-sided DM test proposed by Diebold and Mariano (1995).[14] Although the DM test is robust to the presence of revisions, it does not take into account parameter uncertainty. Consequently, we also implement the AM test of equal predictive ability between two nested quantile regression models when the factors are subject to revisions, as proposed by Amburgey and McCracken (2022). It is important to note that the AM test can be implemented to the $QS(i, \tau)$ statistic but not to the CRPS(i) statistics in (6). Fig. 12 plots the combinations of countries and loss functions for which the equal predictability of the FA-QR and AR-QR models is rejected at a significance level of 10% using the DM test. Consider first the results in Europe. We can observe that the forecast ability of international factors is supported in most countries when using the $QS(i, 0.50)$ and CRPS-R(i) loss functions. However, the null is scarcely rejected when using the $QS(i, 0.05)$ and CRPS-L(i) losses. It seems that international factors are clearly relevant to forecasting upside but not downside inflation risks in Europe. It is remarkable that when comparing these results with those reported in Fig. 3, we can observe that even though the international factors are significant to explain the left quantiles of the inflation density in-sample, they are not significant when forecasting out-of-sample.

Considering now the results for the other three geographical regions, we observe that the combinations of countries and loss functions for which the international factors have forecast power are clearly less than those found in-sample. Still, it is remarkable that there is a large number of countries for which international factors have forecast power, which is not concentrated in the right tail of the inflation distribution but spread over all loss functions. The conclusions are similar when looking at the results of the AM test plotted in Fig. 13, although in this case, it is even harder to reject the null hypothesis.

To summarise the results for all $N = 115$ countries, the upper panel of Table 2 reports the percentage of countries in which the null of the DM test is rejected for the different losses considered. Focusing on the results

[14] Issues associated to testing for equal predictive ability among nested models are avoided by using a rolling-window scheme. In fact, in this context, the test of Giacomini and White (2006) is equivalent to the traditional (Diebold & Mariano, 1995) test; see for example Medeiros et al. (2021).

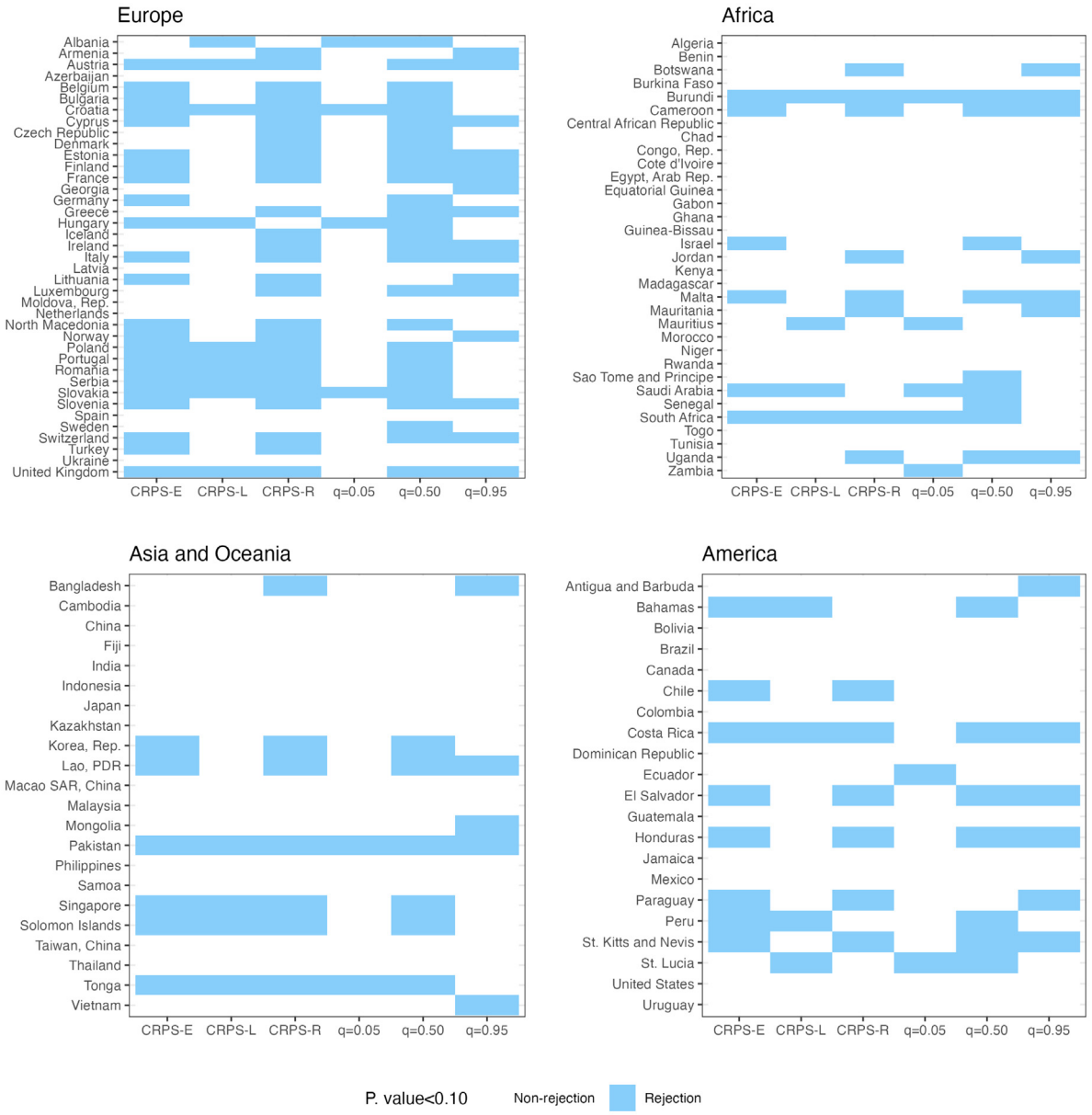


Fig. 12. Tests of significance of equal forecast power of the FA-QR and AR-QR models using the DM test. In blue: rejection of the null in favour of the FA-QR forecasts, when the *p*-value is less than 0.10. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for the CRPS-R loss, we can observe that international factors are relevant for at least one quantile in 43.5% of the total number of countries. This percentage increases to 71.80% among European countries, and to 71.4% among advanced economies. It is also remarkable that the CRPS-R obtained when using the international factors to forecast the quantiles of the inflation distribution is significantly smaller in 23.70% of low-income emerging economies; see Ha, Ivanova, Montiel, and Pedroni (2019) for the influence of international shocks on inflation in low-income economies. The results, summarised in Table 2 for the DM test, clearly point towards the relevance of considering international movements in inflation when forecasting the quantiles of domestic inflation.

It is important to remark that the difference between the results about the predictive content of international factors to forecast domestic inflation quantiles obtained in-sample and those obtained out-of-sample could be due to the smaller number of out-of-sample forecasts. It is well known that small out-of-sample sizes can reduce the ability to detect true predictability; see for example Inoue and Kilian (2005), who show that in-sample tests have higher power than out-of-sample tests and suggest that the results of in-sample predictability will typically be more credible than the results of out-of-sample tests,

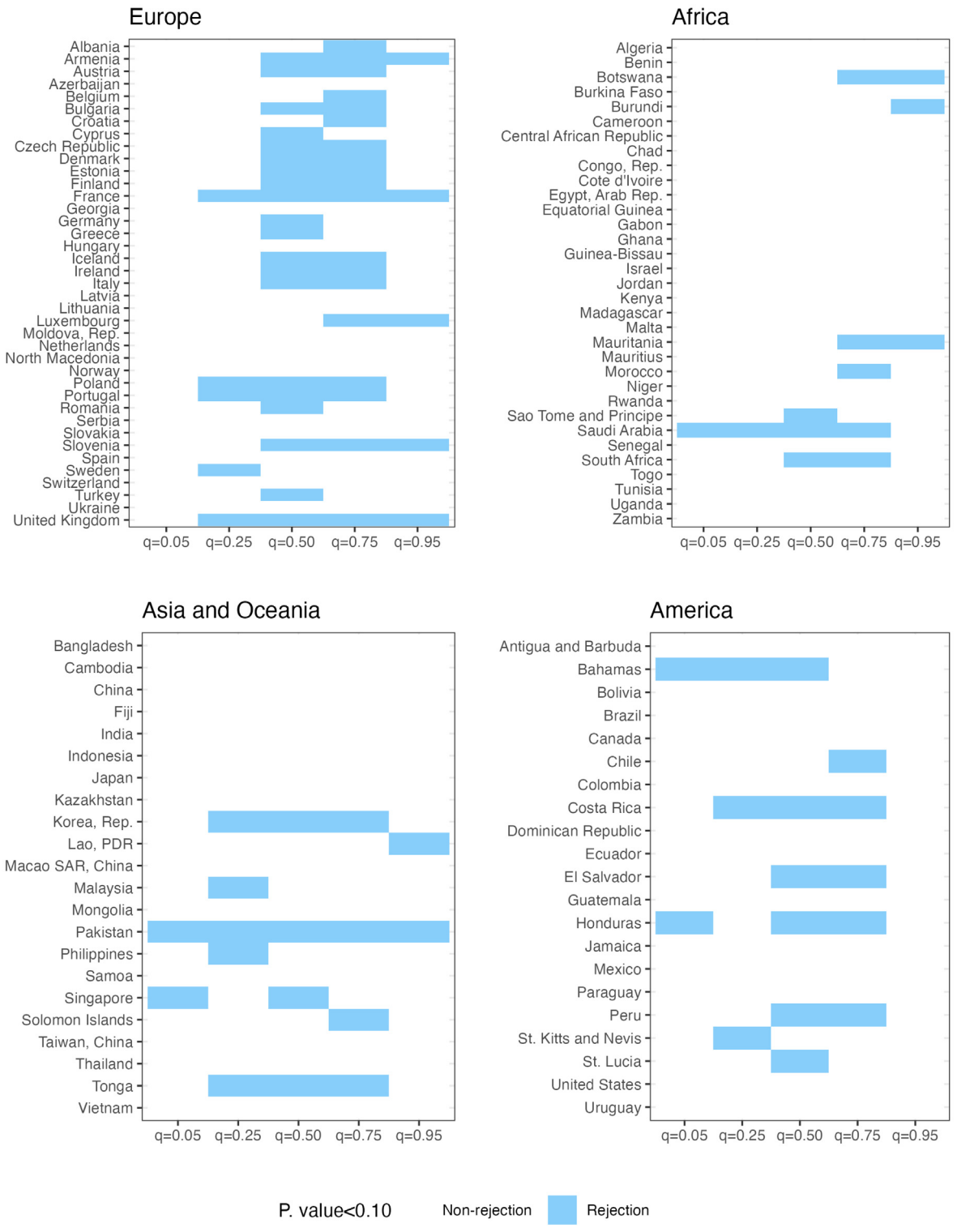


Fig. 13. Tests of significance of equal forecast power of the FA-QR and AR-QR models using the AM test. In blue: rejection of the null in favour of the FA-QR forecasts, when the *p*-value is less than 0.10. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and see Clark and West (2007) for the underrejection of out-of-sample tests.

Also, in order to avoid problems associated with multiple testing when testing for equal predictive ability in

Table 2
Percentage of countries with DM test significant at the 10% level (upper panel) and tests for overall equal predictive accuracy with p-values in parenthesis (lower panel).

	QS(0.05)	QS(0.50)	QS(0.95)	CRPS-E	CRPS-L	CRPS-R
%total	11.3	43.5	29.6	37.4	19.1	42.6
% total in Africa	15.2	27.3	21.2	18.2	12.1	24.2
% total in America	9.5	33.3	28.6	38.1	19.0	28.6
% total in Asia/Oceania	9.1	27.3	22.7	27.3	18.2	31.8
% total in Europe	10.3	71.8	41.0	59.0	25.6	71.8
% total in ADV	5.7	71.4	42.9	57.1	17.1	68.6
% total in HMI	19.0	38.1	19.0	40.5	31.0	35.7
% total in LI	7.9	23.7	28.9	15.8	7.9	26.3
$S_{NP}^{(1)}$	−2.83	8.06	2.79	6.62	2.60	6.53
	(0.00)	(0.00)	(0.01)	(0.00)	(0.01)	(0.00)
$S_{NP}^{(3)}$	−1.43	2.05	1.69	1.81	1.00	2.00
	(0.15)	(0.04)	(0.09)	(0.07)	(0.32)	(0.05)

each of the $N = 115$ countries in the sample, we test for overall predictive ability using the $S_{NP}^{(1)}$ and $S_{NP}^{(3)}$ tests proposed by Akgun et al. (2024), which are based on the sample mean loss differential over the out-of-sample period and across countries, assuming independence and cross-sectional dependence of the loss differentials, respectively. The lower panel of Table 2 reports the p-values of the $S_{NP}^{(1)}$ and $S_{NP}^{(3)}$ tests. Looking at the results for the $S_{NP}^{(1)}$ test, we can observe that the null of overall equal predictive accuracy is rejected in favour of the FA-QR model and, consequently, supports the predictive power of international factors, for all loss functions except $QS(0.05)$. The same conclusion is obtained when looking at the $S_{NP}^{(3)}$ test, although in this case, the null is not rejected for both $QS(0.05)$ and $CRPS-L$. We can conclude that international factors are relevant to predict inflation densities except for the extreme left quantiles.

Finally, we test for equal predictive ability using the fluctuation test of Giacomini and Rossi (2010), which compares the predictive performance of the FA-QR and the benchmark AR-QR model in each moment of time. In order to detect 'pockets of predictability', we run the fluctuation test of Giacomini and Rossi (2010) to compare through time equal predictive accuracy between the FA-QR model and the benchmark AR-QR model. The results are plotted in Fig. 14 for the economies within the G7 and G5. The null of equal predictive accuracy is rejected in the more recent years in the G7 economies, except in the US and Canada. The forecasting power of the international factors increases over time. It is stronger after 2022, and it is mostly related to losses based on the right tail of the distribution of inflation. Consider, for example, the case of domestic inflation in the UK, with the forecasting accuracy of international factors increasing over the out-of-sample period and being significant after 2022 for all losses, except QS(0.05). The picture is somehow different when looking at the G5 economies. We observe that in China, Mexico, and Brazil, the forecasting power of international factors has been approximately constant over the out-of-sample period, while in India, they have been losing power. However, in South Africa, the pattern is similar to that observed in the G7 economies but with the international factors being more important for explaining the left tail than the right tail of the distribution of domestic inflation.

4. Explaining cross-country heterogeneity of the forecasting performance of international factors

To explore the heterogeneity of the results among different economies around the world and the role of different macroeconomic variables in the out-of-sample performance of international factors, we fit the following fixed effects panel regression model:

$$D_{it^*} = \alpha_i + \gamma_{t^*} + \delta_1 O_{it^*} + \delta_2 G_{it^*} + \delta_3 I_{it^*} + \delta_4 U_{it^*} + \delta_5 CBIE_{it^*} + \delta_6 GPR_{it^*} + \epsilon_{it^*}, \quad (7)$$

where, due to some of the explanatory variables being available only annually, t^* is defined in annual terms as $t^* = 2012$ up to 2022. Similarly, because of data availability, the number of countries used to estimate the panel model in (7) is $i = 1, \ldots, 38$.[15] The endogenous variable D_{it^*} measures the relative forecasting performance of the FA-QR model with respect to the benchmark AR-QR model for different loss functions. In fact, for loss function L and months $s(t^*) = 1, \ldots, 12$ belonging to year t^*, the loss differential is defined by $D_{it^*} = \frac{1}{12}\sum_{s(t^*)=1}^{12}(L^A_{is(t^*)} - L^F_{is(t^*)})$, where $L^F_{is(t^*)}$ and $L^A_{is(t^*)}$ are the loss functions of the FA-QR and AR-QR models, respectively, for country i in month s of year t^*. We consider the difference of loss functions for $QS(\tau)$, for $\tau = 0.05, 0.50$, and 0.95, and for $CRPS - E$, $CRPS - L$, and $CRPS - R$. The parameters α_i and γ_{t^*} are the fixed effects for each country and year, respectively. Among the explanatory variables in the panel model in (7) is trade openness O_{it^*}, which, for each year and country, is calculated as the sum of exports and imports divided by GDP; see Attinasi and Balatti (2021) for the relevance of trade integration between countries on the globalisation of inflation. Additionally, G_{it^*} is GDP growth, I_{it^*} is the average inflation rate, and U_{it^*} is the unemployment rate.[16] Furthermore, following Kraskini

[15] Data are available for Belgium, Brazil, Canada, Chile, China, Colombia, Denmark, Egypt-Arab Rep., Finland, France, Germany, Hungary, India, Indonesia, Israel, Italy, Japan, Korea Rep., Malaysia, Mexico, Netherlands, Norway, Peru, the Philippines, Poland, Portugal, Saudi Arabia, South Africa, Spain, Sweden, Switzerland, Taiwan, Thailand, Tunisia, Turkey, Ukraine, the United Kingdom, the United States, and Vietnam. Note that the countries in the sample are spread over the four geographical regions considered and the three groups of economic development.

[16] These data were primarily collected from the World Bank database and national statistics.

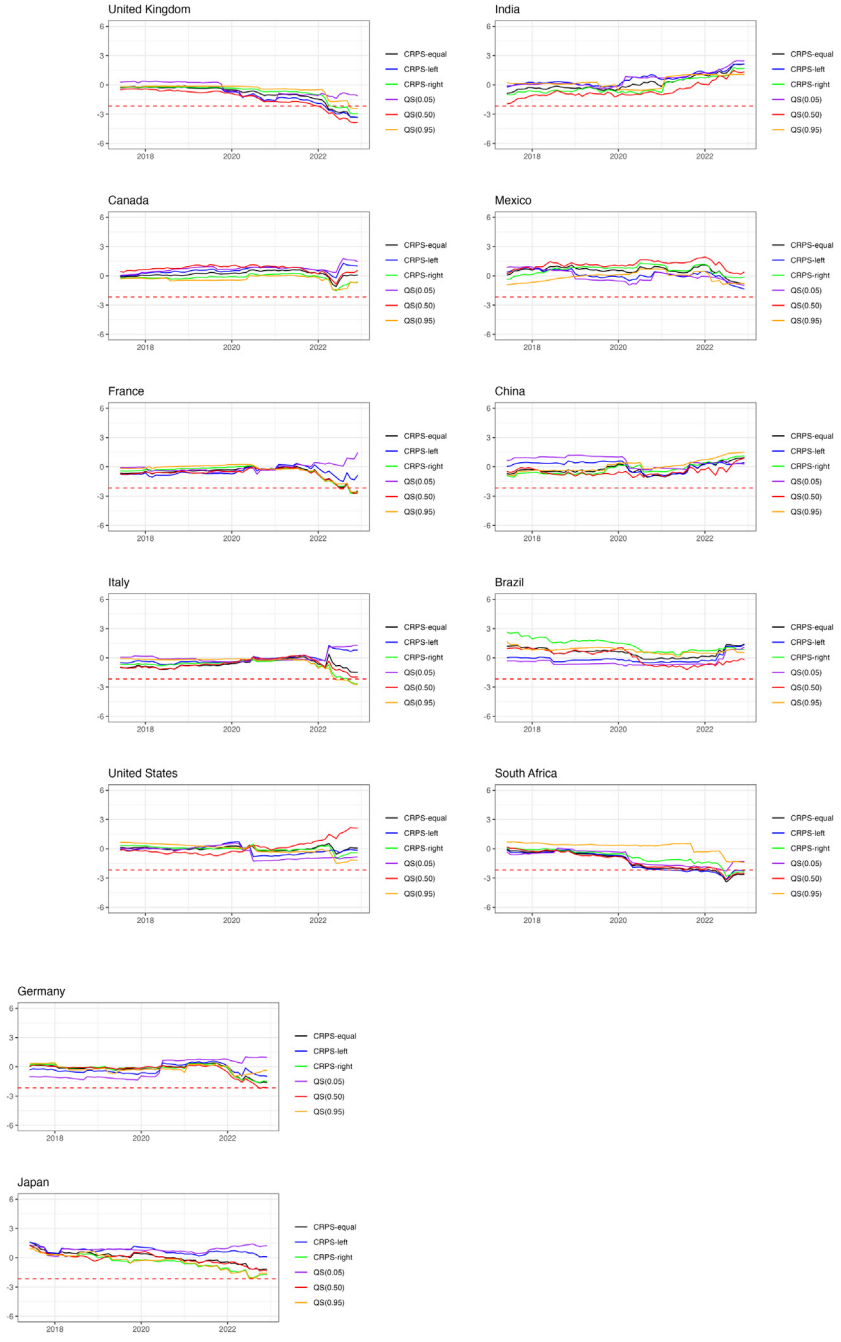


Fig. 14. Fluctuation test for equal predictive ability of the FA-QR model with respect to the benchmark AR-QR model: G7 countries (left column) and G5 countries (right column).

and Gara (2025), we consider the monetary policy independence and geopolitical risk as potential variables explaining the forecast power of international factors. In particular, the monetary policy independence is measured by the Central Bank Independence (CBIE) index proposed by Romelli (2022). The CBIE index is computed as the average of the scores across the following six dimensions: governor and central bank board, monetary policy and conflict resolution, objectives, limitations on lending to the government, financial independence, and reporting and disclosure. The index ranges from 0 to 1, with 0 corresponding to the lowest level of independence and 1 to the highest level. The geopolitical risk (GPR) index is obtained from Caldara and Iacoviello (2022).

Model (7) is estimated as proposed by Beck and Katz (1995), combining ordinary least squares with panel-corrected standard errors to address contemporaneous

Table 3
Estimated parameters of panel model.

	CRPS-E	CRPS-R	CRPS-L	QS(0.05)	QS(0.50)	QS(0.95)
Openness	0.002	0.002**	−0.0004	−0.001	0.002	0.004**
	(0.002)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)
GDP growth	0.007	0.018**	−0.015***	−0.043***	0.001	0.030**
	(0.016)	(0.008)	(0.005)	(0.008)	(0.012)	(0.014)
GDP pc growth	0.016	−0.009	0.021***	0.047***	0.014	−0.020
	(0.015)	(0.008)	(0.005)	(0.007)	(0.012)	(0.013)
Inflation	0.022***	0.007***	0.008***	0.008***	0.010***	0.007**
	(0.003)	(0.002)	(0.001)	(0.002)	(0.003)	(0.003)
Unemployment	−0.0003	−0.005	0.005**	0.010***	−0.0002	−0.007
	(0.008)	(0.004)	(0.003)	(0.004)	(0.006)	(0.007)
GPR	0.071***	0.034***	0.005	−0.025**	0.073***	0.015
	(0.025)	(0.013)	(0.008)	(0.012)	(0.019)	(0.022)
CBIE	−1.685***	−0.600**	−0.595***	−0.900***	−0.844*	−0.737
	(0.585)	(0.295)	(0.184)	(0.280)	(0.450)	(0.515)
Constant	1.196*	0.269	0.567***	0.869***	0.363	0.087
Adjusted R^2	0.265	0.262	0.207	0.271	0.245	0.129

Note: Standard errors in parenthesis.
* The p-value is $p < 0.1$.
** The p-value is $p < 0.05$.
*** The p-value is $p < 0.01$.

correlation across units and unit-level heteroscedasticity. Table 3 reports the ordinary least squares estimates for different forecast loss functions. Several important conclusions can be extracted from these estimates. First, regardless of the loss function considered, the FA-QR model tends to perform better. Consequently, international factors are relevant for forecasting the inflation densities of countries with high inflation rates and small monetary policy independence. Focusing on the right tail of the inflation distribution, i.e., CRPS-right and QS(0.95), we find that trade openness and economic growth are positively related to the forecasting performance of international factors. This result can explain why international factors are strongly relevant for explaining the probability of high inflation in the heavily trade-integrated European countries. However, we can observe that when forecasting the left tail, the forecast gains of international factors are larger in countries with higher per capita growth and more unemployment, but smaller in countries with large GDP growth. Finally, controlling for all other variables in the model, the median of the inflation distribution can be better forecast in countries with higher geopolitical risk.

5. Final considerations

In this paper, we assessed the contribution of international factors to forecast domestic inflation upside/downside risks. We fit FA-QRs to forecast the quantiles of inflation densities in a balanced panel of inflation observed monthly in 115 countries over the period from January 1999 to December 2022 with the international factors extracted from an ML-DFM. We showed that the influence of international factors, measured as common components of a large set of inflations around the world, is relevant to understand the distribution of domestic inflation in a large proportion of countries. These can be global factors (with weights for most countries), regional factors, factors common to economies in a given region, or factors associated with the level of economic development. Looking at the in-sample fit, we showed that, in all but five of the 115 economies considered, the international factors (either the global, regional, or development-level factors) were significant for at least one quantile (usually the upper quantiles in advanced economies and the lower quantiles for developing economies). We also showed that the relevance of international factors is stronger in advanced than in low-income economies. We analysed the out-of-sample forecast power of the international factors with similar conclusions, although the international factors were less significant. The observed heterogeneity among countries can be explained by inflation also depending on domestic drivers within each country; see for example Busetti et al. (2021) and Binici et al. (2024), who analysed and disentangled the confluence of domestic and external factors in explaining the evolution of inflation dynamics in Europe, where post-pandemic inflation reached the highest level in four decades. After analysing whether the forecasting power of international factors depended on domestic variables, we showed that the level of inflation, GDP growth, per capita GDP growth, unemployment, geopolitical risk, trade openness, and central bank independence were variables that explained the heterogeneity of the results.

Our results can help to understand better some of the puzzles about inflation modelling and forecasting, which are often related to domestic models; see for example the very recent essay by Mankiw (2024). We have seen that international factors should also be considered for an appropriate understanding of domestic inflation in a large proportion of countries.

The increasing influence of international factors on domestic inflation in advanced economies points towards the coordination of corrective actions by central banks. It seems important that central banks in developed countries coordinate their strategies to control the risk of high domestic inflation. Furthermore, if inflation is largely determined abroad, a central bank could be less concerned about inflation exceeding its target, prioritising job creation. Alternatively, it may also need to make larger

adjustments to interest rates to stabilise inflation; see Forbes (2020). Due to the increasingly influential international factors, Ha et al. (2022) pointed out that central banks in EMD economies face more challenges than those in advanced economies. They recommend a careful calibration, credible formulation, and clear communication of policies in EMD economies to make them more resilient to international factors of inflation.

CRediT authorship contribution statement

Ignacio Garrón: Writing – review & editing, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vladimir Rodríguez-Caballero:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Esther Ruiz:** Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.12.003.

Data and code availability

The code and data necessary to reproduce the results gathered in this manuscript are available at https://github.com/IgnacioGarron/replication_ijf_int_inflation_factors_risks.

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